The Noisy Work of Uncertainty Visualisation

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Abstract

Better representation of the uncertainty in a data visualisation is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion results in a significant amount of conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarise the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. It is hoped that it will be useful for guiding new graphics methodology and experimental research.

1 Introduction

What do we mean when we talk about "uncertainty visualisation"? The phrase can feel like a bit of a contradiction to anyone who is familiar with the term. Among statisticians, "uncertainty" is often discussed as though it is an omnipresent spectre that touches every stage of our analysis without ever being fully seen. Authors will often mention that the phrase is incredibly vague (Spiegelhalter 2017, Griethe & Schumann 2006), or talk around the topic by describing a list of things uncertainty *could* be (Kinkeldey et al. 2014, Hullman 2016), but rarely do we see authors attempt to discuss what uncertainty actually *is*. Almost as if in direct contrast with this, visual statistics (information visualisations, data plots) are one of the most powerful tools in the statistician's toolbox, allowing for quick and memorable communication that identifies quirks in our data that we didn't even know to look for. We see this in datasets such as Anscombe's quartet (Anscombe 1973) or the Datasaurus Dozen (Matejka & Fitzmaurice 2017, Locke & D'Agostino McGowan 2018), where visual statistics are able to highlight elements of the data that are invisible to the typical summary statistics. We also see this in recall experiments, where something as simple as sketching a distribution before recalling statistics or making predictions can greatly increase the accuracy of those measures [Hullman et al. (2018); Goldstein2014]. Taken together, uncertainty visualisation implies a need to pull back the curtain and expose put light on the unknowns of our analysis.

As nice as this sentiment is, it turns out to be easier said than done. Reviews on uncertainty visualisation rarely offer tried and tested rules for effective uncertainty visualisation, but rather, they comment on the difficulties faced when trying to summarise the field. Kinkeldey et al. (2014) found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or a greater goal of describing general principles. Hullman (2016) noticed there is a serious noise issue in the field, with noise from participants misun-

derstanding visualisations, misinterpreting questions, and incorrectly applying heuristics, all overwhelming any information we can glean from studies. MacEachren et al. (2005) identified so much contradicting evidence that they were able to spend an entire page discussing the conflicting evidence for the question: "Should I map uncertainty to colour hue?" Spiegelhalter (2017) concluded that different plots are good for different things, and ultimately disagreed with the goal of identifying a universal best plot for all people and circumstances. Padilla, Kay & Hullman (2022) summarised several cognitive effects that repeatedly arise in uncertainty visualisation experiments; these effects were each discussed in isolation as a list of considerations an author might make, rather than an overarching theory of rules for effective uncertainty visualisation.

While these reviews are thorough in scope, none discuss how the existing literature contributes to the broader goal of uncertainty visualisation. The problem faced by the literature is easily summarised with a famous quote by Henri Poincaré.

> "Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house." - Henri Poincaré (1905)

That is to say, despite the wealth of reviews, the field of uncertainty visualisation remains a heap of stones. There is a mountain of work that identifies common heuristics found in uncertainty visualisations, evaluates competing plot designs, or starts a theoretical discussion on a niche aspect of the field. This is important work that needs to be done, but each of these papers offers up its own bespoke motivation and methodology, with little reference to the uncertainty visualisation papers outside their periphery. This becomes even more difficult to manage when these studies conflict. The field is in desperate need of a unifying theory that can tie this swath of research together. This review attempts to address this

issue by offering a novel perspective on the uncertainty visualisation problem. That is, we are going to use the wealth of established stone to construct the foundations on which we can build a house.

2 The purpose of uncertainty visualisation

Mentions of "uncertainty visualisation" start springing up around 1990, across several different fields (Ibrekk & Morgan 1987, MacEachren 1992), each with its own motivation for the work. In computer science the area appears to be motivated by issues in the public's perception of random variables, with the hope that visualisations would give laypeople the ability to extract important information from graphical representations (Ibrekk & Morgan 1987). With similar concerns about the public's understanding of randomness, the field of psychology, statistics, and economics used "uncertainty visualisations" as a communication tool to mitigate the psychological bias associated with the communication of risk, a topic of concern since the early 1980s (Spiegelhalter 2017). In the cartography literature, it was motivated by inherent uncertainty of geoscience data, the practical use of visualisation as an exploratory tool, and the constrained visual channels from map representations (MacEachren 1992). These disparate motivations have merged together, and today, uncertainty visualisation is usually motivated by the vague goal of "decision-making". This term has been used to mean the mitigation of psychological bias to ensure economically rational decisions (Padilla, Hosseinpour, Fygenson, Howell, Chunara & Bertini 2022, Padilla et al. 2021, Kale et al. 2021), the facilitation of trust or confidence (Zhao et al. 2023, Yang et al. 2023), the ability to extract values related a distribution (Sarma et al. 2023), the prevention of false discovery in plots (Sarma et al. 2024, Koonchanok et al. 2023); or the extraction of some other metric that is vaguely related to "uncertainty" (Chakraborty et al.

2024, Ndlovu et al. 2023). This gradual scope creep of the field, motivated by the hazy definition of "decision-making", is the most likely culprit for the irreconcilable literature that makes up the field today.

Given that there is so much subliminal disagreement in uncertainty visualisation, how did these disparate motivations come to be seen as interchangeable? All discussions on uncertainty visualisation seem to have a common thread that connects them; the belief that the ultimate goal of uncertainty visualisation is not trust, rationality, or value extraction, but *transparency*. We see it said directly in reviews of the field (Padilla, Kay & Hullman 2022). We see it when authors claim that failing to include uncertainty is akin to fraud or lying (Hullman 2020, Manski 2020). We see it when authors assert that uncertainty communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference (Correll & Gleicher 2014, Kale et al. 2018, Griethe & Schumann 2006). We see it when authors say uncertainty visualisations should communicate a degree of confidence (Boukhelifa et al. 2012) or validity (Hullman 2020, Griethe & Schumann 2006) in our conclusions. We see it when authors suggest uncertainty visualisation should "guide, qualify, or soften our judgements of uncertain data" (e.g. Wilkinson (2005) in his seminal work on the grammar of graphics). These authors are not wrong about the need for transparency in science communication, either. A six-month survey of anti-mask groups on Facebook during the COVID-19 pandemic showed that anti-maskers made persuasive arguments by exploiting inherent uncertainty that was ignored by pro-maskers (Lee et al. 2021).

This consensus leads us to understand that uncertainty visualisation is motivated by the need for a sort of "visual hypothesis test", a sentiment expressed by some authors directly (Correll et al. 2018, MacEachren 1992). A successful uncertainty visualisation would act as a "statistical hedge" for any inference we make using the graphic. Since the purpose of a

visualisation is to give a quick gist of the information (Spiegelhalter 2017), this hedging should be communicated visually without the need for complicated mental calculations. Therefore, an effective uncertainty visualisation should not just "show" uncertainty; untrustworthy conclusions should *not be visible*. If we refer to the conclusion we draw from a graphic as its signal and the uncertainty that should make this signal harder to read as the "noise" increases, we can summarise the above information into three key requirements. A good uncertainty visualisation should:

1) Reinforce justified signals to encourage confidence in results.
2) Hide spurious signals that are overwhelmed by noise.
3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.

Usually, visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals. Therefore, we **suggest calling this approach to uncertainty visualisation "signal-suppression"** since it primarily differentiates itself from the normal "noiseless" visualisation approach through criterion (2). This is the main criterion we will use to assess the current literature on uncertainty visualisation.

3 Current Approaches

3.1 Ignoring uncertainty

The most common way to visualise uncertainty is to simply not do it at all. A study conducted by Hullman (2020) found that only a quarter of authors surveyed included uncertainty in 50% or more of their visualisations. Part of the abstinence comes from the

fact that authors are not sure how to calculate the uncertainty (Hullman 2020). This is not entirely unreasonable, given that even visualisation authors themselves seem to be in conflict about what exactly uncertainty is. Therefore, it makes sense to start with visualisations that ignore uncertainty with the hope that, by looking at where uncertainty *isn't*, we can better understand where it *is*. (XXX don't understand the secon part of this statement)

3.1.1 What is uncertainty?

It is surprisingly hard to describe what uncertainty actually is. Most authors simply avoid the problem and describe the characteristics of uncertainty, of which there are plenty. Often, uncertainty is split using an endless stream of ever changing boundaries, such as whether the uncertainty is: due to true randomness or a lack of knowledge (Begg et al. 2014, Spiegelhalter 2017, Gustafson & Rice 2019, Padilla, Kay & Hullman 2022, Hullman 2016, Walker et al. 2003) quantifiable or unquantifiable (Spiegelhalter 2017, Walker et al. 2003, Padilla, Kay & Hullman 2022); scientific or human (Benjamin & Budescu 2018, Gustafson & Rice 2019); systematic or random (Sanyal et al. 2009); statistical or bounded (Gschwandtner et al. 2016, Olston & Mackinlay 2002); accuracy or precision (Griethe & Schumann 2006, Benjamin & Budescu 2018, Hullman 2016); etc. There are enough qualitative descriptors of uncertainty to fill a paper, but none of this is particularly helpful in understanding how to integrate it into a visualisation.

Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*. Descriptive statistics describe our sample as it is and summarise large data down into an easy-to-swallow format. Descriptive statistics are not seen as the primary goal of modern statistics; however, this was not always the case. In 19th-century England, *positivism* was the popular philosophical

approach to science (positivists included famous statisticians such as Francis Galton and Karl Pearson). Practitioners of the approach believed statistics ended with descriptive statistics as science must be based on actual experience and observations (Otsuka 2023). In order to make statements about population statistics, future values, or new observations, we need to perform inference, which requires the assumption of the "uniformity of nature", that is, we need to assume that unobserved phenomena should be similar to observed phenomena (Otsuka 2023). Positivists believed referencing the unobservable was bad science. In other words, these scientists embraced descriptive statistics due to the inherent certainty that came with them. Since uncertainty is nonexistent in descriptive statistics, it is clear that uncertainty is a by-product of inference. Uncertainty is the noise that is both inseparable from our inference and meaningless without it.

If we consider uncertainty as a byproduct of statistical inference, it would make sense to simply quantify it as a distribution. Suggesting distributions as a representation of uncertainty is not new. Kay (2023) originally suggested thinking about uncertainty visualisations as visualisations with distribution inputs, to replace the commonly used mean and standard deviation, removing the assumption of a Gaussian distribution. Taking this point a step further, we suggest that a distribution is sufficient to characterise *any* uncertainty, no matter how intangible it is. This is in contrast to the assertions made by some authors, who believe abstract notions of uncertainty, such as credibility (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or uncertainty about uncertainty (Hadjimichael et al. 2024), are too complex to be quantified. This is simply untrue, as abstract concepts such as human belief or credibility are regularly quantified by Bayesians and hierarchical models are often used for uncertainty about uncertainty.

3.1.2 Example: ignoring uncertainty

If visualising uncertainty is fundamentally a scenario where we are visualising a set of random variables, what does "ignoring" uncertainty look like? Figure 1 shows plots of data from three different scenarios, each with either high or low uncertainty. The plots show the expected value of the input distribution. In plots R1 and R2, the expected value is the line from a simple linear regression on the `mtcars` data (available in `ggplot2` (Wickham 2010), and originally from Henderson & Velleman (1981)), with differing sample size underlying the uncertainty. Plots S1 and S2 show a choropleth map of Iowa, where counties are coloured according to a simulated temperature measurement, with measurement error being the uncertainty associated with the measuring instrument. Plots G1 and G2 show samples simulated from five populations (A-E). In G1 distributions have the same low variance, and in G2 they have the same high variance. In the linear regression, we see a downward trend, and because this is overlaid on the observed data we can get a sense of the uncertainty. It is not entirely ignored in this plot, because the data is plotted. However, the uncertainty of interest is in relation to the regression line. In the map you should see a sine wave spatial trend. In the univariate distributions, an incremental increase in treatment. If we were to ask a reader: Can you see a difference between the plots in the top row versus the bottom row? Is the strength of the trend communicated through the visualisation? The answer to both of these questions will be no, for S1, S2, G1, G2, as the high and low variance cases are identical. For R1 and R2, the answer would likely only relate to the points in the plots, not the regression line.

We will use these examples as touchstones for this discussion on uncertainty visualisation, particularly to discuss the approaches suggested in the literature.

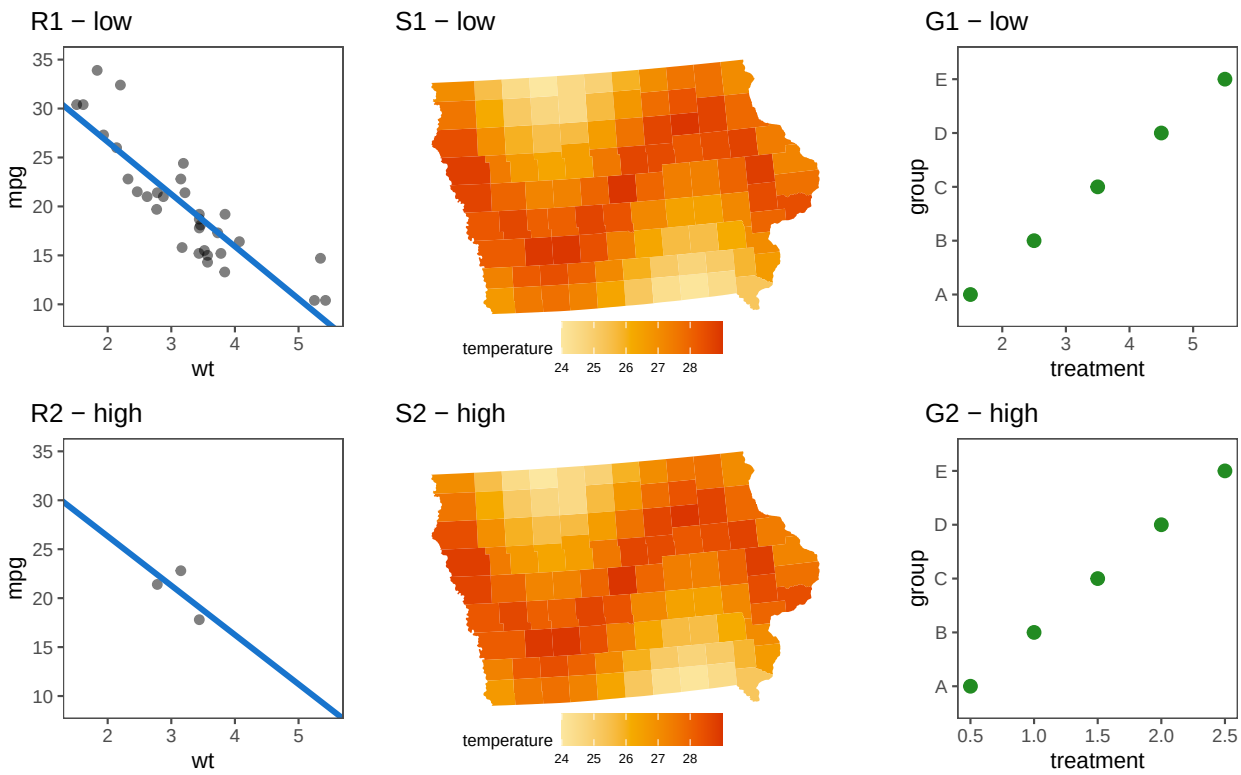

Figure 1: Three example types of data and associated plots that will be used to illustrate various choices of uncertainty representation throughout the paper: scatterplot and regression line (R1, R2), spatial choropleth map (S1, S2), grouped dotplot (G1, G2). Two levels of uncertainty (low, high) is used with each example. Only with plots R1 and R2, where the regression overlays the scatterplot, can we see a difference between low and high uncertainty. Ignoring uncertainty can lead to misrepresentation of data.

3.2 Uncertainty as a statistic

Uncertainty is often treated as another statistic. An example is an exceedance probability map of spatial data (see Kuhnert et al. (2018) with software available in Lucchesi et al. (2021)). Here probability of exceeding a specified value is displayed on the map. This places the focus on extreme events. Another is the icon array used for risk communication (e.g. Spiegelhalter (2017)). Here the probability is translated to a set of icons, like people images, coloured for happening or not. It is argued that people will understand the impact

if they can see the coarse number of affected individuals. The summary plot, proposed by Potter et al. (2010) developed to address concerns about uncertainty in data displays, is an extraordinarily dense chart. It combines the plot of a single set of numeric values, with a density estimate, boxplot elements and statistical moments.

Some authors take this approach because they explicitly believe uncertainty is a variable of importance in and of itself (Blenkinsop et al. 2000), others straddle the line, asserting uncertainty is acting as signal and noise and should fulfil both roles (Peña-Araya et al. 2025). Simply put, these approaches might be described as swapping out the statistic for an "uncertainty statistic" to get an "uncertainty visualisation". Is it really that easy?

3.2.1 Example: visualising variance

Figure 2 depicts the six plots showing this approach for the data introduced in Figure 1. The original central value estimate has been replaced with an uncertainty statistic. Plots R1, R2 are absurd examples of this approach: the regression line replaces the estimated slope and intercept of a line with their standard errors. To understand to plot you would read: the intercept of the line now says that the standard error of the trend is close to 0 in R1 and 20 in R2, and the error of the slope is low for R1 and very high for R2. Sometimes this approach is sensible, as we can see in the exceedance probability maps depicted in plots S1 and S2. These map $P(temperature > 27)$ to colour for each county. The sine wave trend is clearly visible when error is low but barely visible with high error. This does reveal something interesting: that the variance is constant in G1 but different for eacg group in G2. It is a nonsensical display, though, because the trned has completely disappeared. If this approach is used in practice, you tend to see two plots, one focusing on the main estimate and the other on the uncertainty.

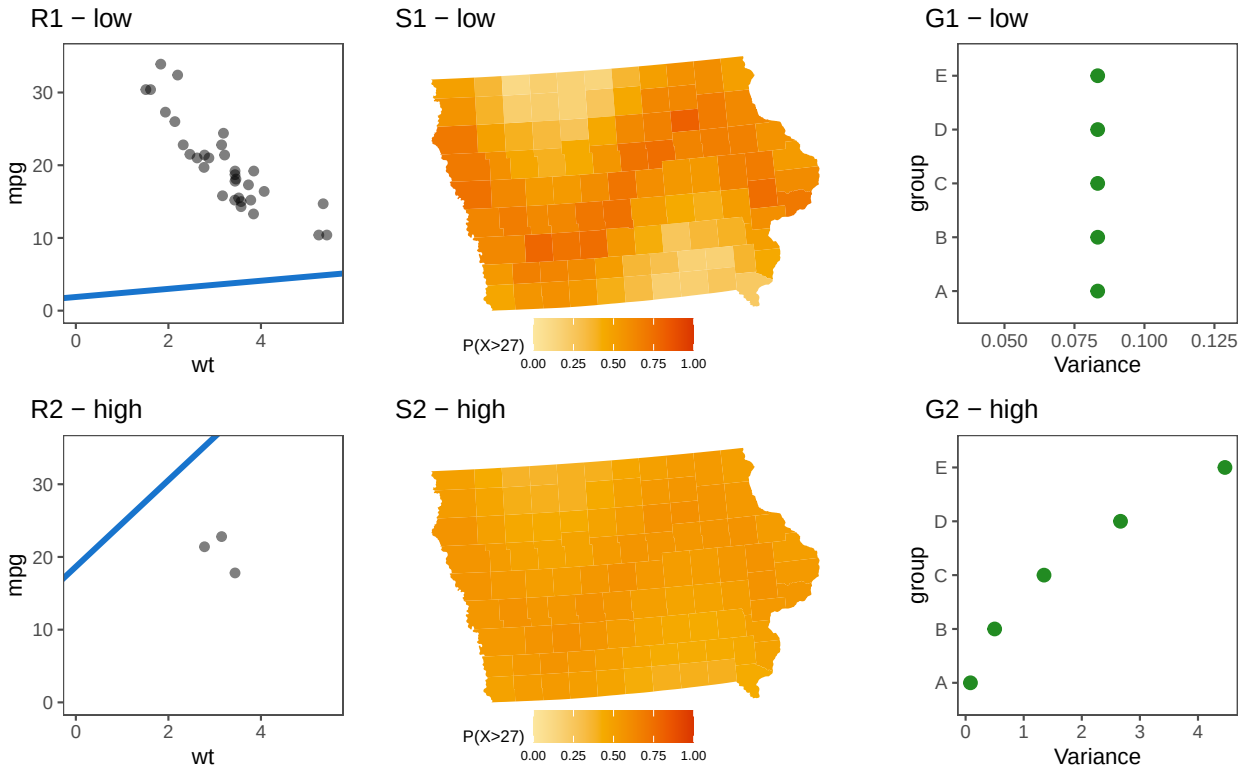

Figure 2: Treating the uncertainty as a statistic, to the extreme, with the same six examples. The easiest to digest is the choropleth (S1, S2): instead of temperature, the probability of exceeding 27°C is shown. This has the effect of highlighting (sine wave) trend in the low error data, and de-emphasizing it in the high error data. The other four take some thinking. In R1, R2, the line now indicates the standard error of the intercept and slope of the regression line, and one would read that the error for 'R1 - low' is close to 0 but 'R2 - high' is closer to 20, and similarly for slope. Yes, this is an absurd display! With G1 and G2, it is similar, but we learn something interesting that was not seen earlier: the variance in G2 is not uniform like that in G1.

3.2.2 What is an uncertainty visualisation, then?

What is, or is not, an uncertainty visualisation is one of the most pervasive divides in the literature. For example, Wilkinson (2005) mentions that popular graphics, such as pie charts and bar charts, omit uncertainty. Wickham & Hofmann (2011) suggests their product plot framework, for area plots like bar charts, and also histograms, needs to be extended to include uncertainty representation. However, pie charts, bar charts and histograms have all been used in a significant number of experiments as examples of an "uncertainty visualisation" (Ibrekk & Morgan 1987, Olston & Mackinlay 2002, Zhao et al. 2023, Hofmann et al. 2012). What is going on here?

This conflict relates to the subconscious disagreement in the purpose of uncertainty visualisation. If you believe uncertainty visualisation is about communicating risks or random variables, uncertainty visualisations are just visualisations of "uncertainty statistics" or distributions. On the other hand, if you believe uncertainty visualisation is about suppressing false signals in a graphic, then you see an uncertainty visualisation as a transformation of an existing graphic that somehow adds the uncertainty in. The former has no limitation on the visual appearance of an "uncertainty visualisation", allowing pie charts, bar charts or histograms, so long as the graphic is visualising "uncertainty", while the latter believes uncertainty visualisations only exist in relation to some "normal" visualisation. When we refer to the graphics depicted in Figure 2 as "uncertainty visualisations", we are classifying visualisations by the data they display, not their visual features. This is not the standard approach in statistical graphics. A scatter plot that compares means and a scatter plot that compares variances are both scatter plots.

Unlike plots, which are not defined by a statistic, uncertainty is *only* defined in relation to our visual statistic, a topic that frequently appears in the literature to be dependent

on the "goals" of our analysis. Meng (2014) commented that what is kept as data and what is tossed away is determined by the motivation of an analysis - what was previously noise can become signal depending on the question to answer. Otsuka (2023) suggested that, the process of observing data to perform statistics is largely dependent on our goals because the process of boiling real-world entities down into probabilities depends on the relationship we seek to identify with our data. Wallsten et al. (1997) argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision-maker wants to represent and why it matters. Fischhoff & Davis (2014) suggested we should have methods for communicating uncertainty depending on what the user is supposed to do with it. Spiegelhalter (2017) says we "cannot assess the quality of risk communication unless the objectives are clear". Peña-Araya et al. (2025) asserted that whether or not it is a source of doubt depends on the context. The sentiment behind this repeated point is clear: the role of uncertainty or signal is not dependent on the "type" of statistic, on the source of the information, or the methods we use; it is determined by the statistic we wish to draw inference on. Therefore, the fundamental problem with the "uncertainty statistic" approach is that the uncertainty in the plot isn't acting as noise; it is acting as signal.

If the uncertainty in a graphic is acting as a signal, there isn't an interesting perceptual challenge associated with the visualisation, and they should just be visualised using current principles of graphic design. In changing the inferential statistic, we also haven't dealt with the original problem of integrating noise, as these "uncertainty statistics" *also have associated uncertainty in the estimates* (e.g. variance of standard deviation estimate) that is being ignored. There is nothing wrong with explicitly visualising variance, error, bias, or any other statistic. Just like any other statistic, these metrics provide important and useful

information for analysis and decisions. The problem with this approach is that it means everything is an uncertainty visualisation, and if everything is an uncertainty visualisation, nothing is.

3.3 Uncertainty as a variable

Another common sentiment in the literature is the characterisation of uncertainty as just another variable that needs to be integrated into the visualisation, which means uncertainty visualisation is, at its core, a high-dimensional visualisation problem (Griethe & Schumann 2006, Peña-Araya et al. 2025). There are examples of this thinking across computer science (Kinkeldey et al. 2014), cartography (MacEachren et al. 2005), and statistical graphics (Wilkinson 2005).

Discussion of the visualisations focus on how "integrated" the uncertainty is with the estimate. Kinkeldey et al. (2014) identified a split between intrinsic plots, where we map uncertainty to the colour or size of the geometric object of our estimate, and extrinsic plots, where uncertainty is mapped to a separate geometric object, such as glyphs or error bars. A similar distinction was made by Padilla, Kay & Hullman (2022), who classified uncertainty visualisations as graphical annotations (extrinsic), and probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two. It is unclear if these levels of "integration" in a plot design affect its ability to do signal suppression.

3.3.1 Example: mapping two independent variables

Figure 3 shows the six examples introduced in Figure 1. Here, uncertainty is mapped to a spare aesthetic in the plot. (In the grammar of graphics, variables are mapped to aesthetics, like position, colour, size, in a plot.) In plots R1 and R2, the standard error of the slope

is represented by the width of the line. This is a a common practice but it fails to also represent the standard error of the intercept. We might think we can examine the width of the line at `wt=0`, but this would be incorrect. Here it has been included, less obviously, by mapping the standard error of the intercept to transparency. While these help to give a gestalt of the uncertainty, they are not exacting representations. One would expect that the width above and below the line is one standard error, but why not map the width to two standard deviations, or even three? Interpreting transparency into a numerical quantity is also virtually impossible, so using this aesthetic mapping is effectively useless.

A bivariate colour palette map is shown in plots S1, S2. With two dimensions of the plot reserved for spatial position, how to incorporate the error has been uncertain itself. The bivariate colour palette maps the estimate to hue and the error to saturation, keeping the signal and noise contained to one visual aesthetic. This has the unfortunate effect of making the signal appear stronger when the error is higher (S2). Perception of colour is a wild beast, hard to tame.

The extrinsic approach, shown in plots G1 and G2, has the estimate represented by a point, and uncertainty computed as a 95% confidence interval mapped to line length. The trend is still visible in both displays. It could be argued that where the variance is high, the trend remains a main focus, which means the display fails to sufficiently suppress the signal.

Because all of these graphics visualise the distribution's estimate and uncertainty as two separate pieces of information, the message of the plot is "here is the trend *and* here is the uncertainty". It is worthwhile to interrogate why this occurs to see if we can move towards a version of this plot where we are able to communicate signal and noise as a single item.

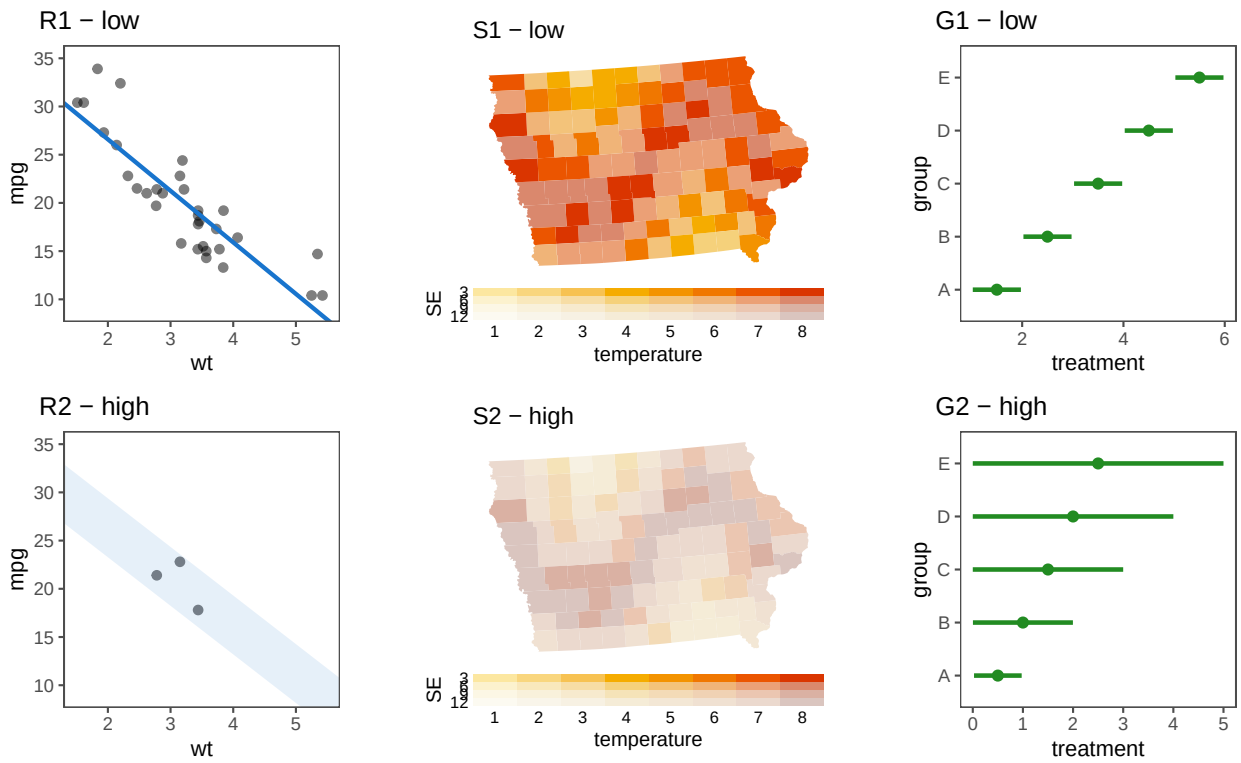

Figure 3: Treating the uncertainty as a variable, with the same six examples. In plots R1, R2 the standard error of the slope mapped to the line width. The standard error of the intercept is mapped to the transparency, which is less conventional. In plots S1, S2 a bivariate colour palette is used with mean mapped to the hue, and error mapped to saturation. Plots G1 and G2 represent the mean of each group as a point, and variance as an interval. The uncertainty is integrated well with the signal, but for the choropleth the result is undesirable: the signal is easier to see when the error is higher.

3.3.2 Can we visualise a "single integrated uncertain value"?

The reality is, based on our discussion on inferential statistics, uncertainty *isn't* a separate variable; it is a component of the random variable that is indistinguishable from the random variable itself. Similarities between the "as a variable" approach and the "as a statistic" approach are apparent when we read motivations for visualising an estimate (i.e. Figure 1) and variance (i.e. Figure 2) side-by-side using two separate graphics. (This organisation could also be considered to suffer from the condition of change blindness (Simons & Levin 1997) as one needs to switch focus between two displays.) The preference for the methods utilised in Figure 3 is usually motivated by the difficulties in combining information on two separate graphics (Moritz et al. 2017, Correll et al. 2018), rather than an understanding that the "uncertainty statistic" approach is not philosophically sound. To achieve signal suppression, we need to visualise noise and signal together as a "single integrated uncertain value" (Kinkeldey et al. 2014) rather than as two separate statistics.

Just because we can still see the signal in Figure 3, that does not mean the reading of the estimate is completely independent of the uncertainty. When making any visualisations, we usually want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others (Smart & Szafir 2019). It is also interesting to know whether readers can see all structures when there are multiple structures present in a data plot, or whether they systematically fixate one one (VanderPlas & Hofmann n.d.). While aiming for separability is useful for standard data visualisation, in uncertainty visualisation it allows the estimate and its variance to be read independently, potentially leading to the uncertainty being ignored (Padilla, Kay & Hullman 2022). Therefore, rather than trying to maintain visual separability, the goals of uncertainty visualisation align far better with the pursuit of visual integrability. In an ideal system, our estimate and

uncertainty would be separately manipulatable, but so *well-integrated that they are read as a single channel by the human brain*. The problem is that even if we can implement the most extreme versions of integrable, our methods fall short, as illustrated by the bivariate colour palette map in Figure 3. Colour hue and brightness are one of the classic examples of integrable variables (Vanderplas et al. 2020), and decreasing saturation should make the colours harder to distinguish, but the signal is still clearly visible with the high variance. This is to say nothing of the fact that multi-dimensional colour palettes can make the graphics harder to read and hurt the accessibility of the plots (Vanderplas & Hofmann 2015).

3.3.3 Another example: mapping combined variables

The Value Suppressing Uncertainty Palette (VSUP) (Correll et al. 2018) were designed with the intention of preventing high uncertainty values from being extracted from a map by blending colours together as they become less certain. Figure 4 shows the spatial example (S1, S2) using the VSUP approach. Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks (Correll et al. 2018) or search tasks (Ndlovu et al. 2023), we can see that, at least for our example, when the uncertainty is high, the spatial trend has functionally disappeared.

Finally, we have signal suppression! Well, not really, sorry, we tricked you. The two plots depicted in Figure 4 actually show the exact same data; they are both the low variance case. If you look closely, you can see that the two plots have a different scale, where plot (a) has been scaled according to our existing knowledge about this data, while plot (b) has been scaled using the range of the data passed to the plot. Since the variance and estimate are scaled independently, arbitrary differences in the range of our variance, unrelated to the

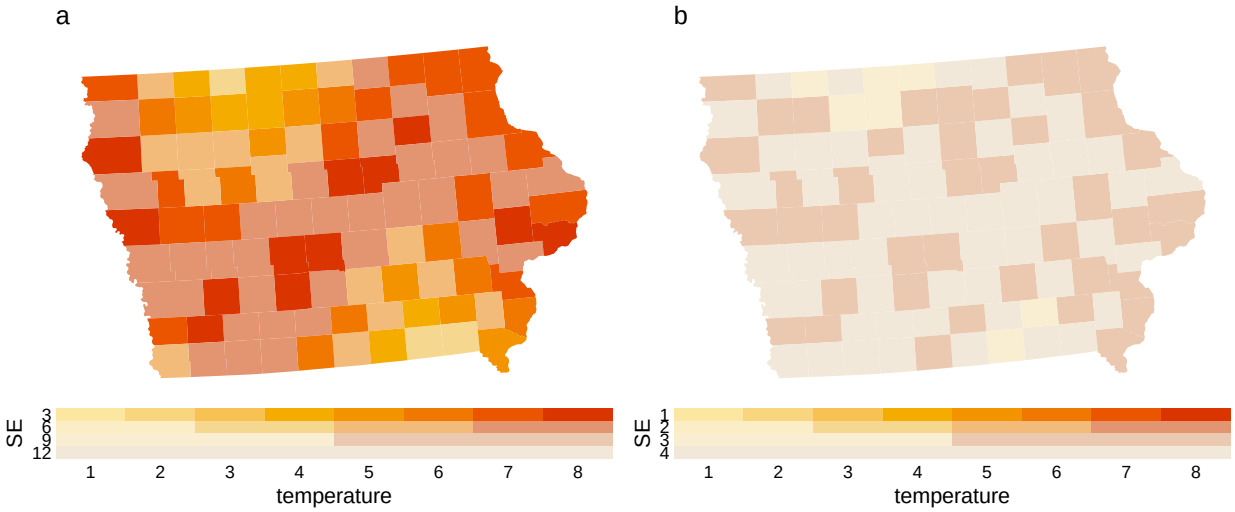

Figure 4: The spatial examples displayed with a choropleth map using a VSUP colour palette, where hue is blended when increased uncertainty. Plot b has successfully produced signal suppression. Look closely at the scales, though: it may have you question whether this should have happened.

estimate itself, will have significant impacts on the visual appearance of our plot. The scale issue in VSUP maps was also recognised by Kay (2019), who noted that the suppression of any one hypothesis largely depends on the methods we use to combine the palette, and the variance levels at which the blending occurs. This means that, for us to know that our plot will successfully perform signal suppression, we need to already know what signal we are trying to suppress and set up the VSUP palette accordingly. This means that VSUP maps are not suitable for exploratory data analysis.

3.3.4 Uncertainty and exploratory data analysis

The reason there is no uncertainty in descriptive statistics is not due to a unique property of descriptive statistics, but rather due to the lack of inference. Descriptive statistics are actually a small piece of a much larger field, exploratory data analysis (EDA), that tends not perform statistical inference. Tukey et al. (1977) described EDA as the process of searching

for interesting hypotheses ("the greatest value of a picture is when it forces us to notice what we never expected to see"), and defined it in relation to confirmatory data analysis (CDA), the process of verifying a hypothesis. There are more subfields of EDA: initial data analysis (Huebner et al. 2016, Chatfield 1985) which involves checking assumption and data quality prior to CDA, and model diagnostics (e.g. Belsley et al. (1980)) which includes posterior checks on the model fit. What binds these pursuits together is their reliance on visual summaries for making assessments, and an absence of formal inference.

Some have argued that the EDA and CDA are not entirely distinct, as it is often difficult to draw a hard line (e.g. Hullman & Gelman (2021)). Our belief, as with many concepts, is that these approaches exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached. It can help to think of the knowledge-generating process of EDA and CDA as the nozzle on a hose with multiple spray options, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

Viewing EDA and CDA as a dichotomy can create some confusion when it comes to understanding the source of uncertainty in our analysis. This is why we have avoided the topic until now, despite the fact that an uncertainty visualisation system for EDA is one of the most discussed topics in the field (Hadjimichael et al. 2024, Peña-Araya et al. 2025, MacEachren et al. 2005, Sarma et al. 2024, Griethe & Schumann 2006). The EDA versus CDA dichotomy can be compared to the dichotomy between induction, for building theories, and deduction, for testing theories, from formal logic. One of the trade-offs in the two methods is that deductive conclusions provide certainty, while conclusions from induction are inherently uncertain. This means that EDA, the inductive counterpart, has

uncertainty in any conclusions reached through the method, with the requirement to follow up our newfound hypothesis with CDA if we want true certainty. This adds another layer of confusion to the study of uncertainty visualisation: conflating the uncertainty in our data with the uncertainty that is inherent to an exploratory process characterised by EDA.

Authors often over-compensate for the inherent EDA uncertainty, to pre-emptively hedge against every false inference that could possibly be drawn from a graphic. Hullman & Gelman (2021) argues there is no such thing as a "model-free" visualisation. Guo et al. (2025) provides a grammar for visualising statistical model checks. The lineup protocol (Buja et al. 2009) provide the viewer with plots of the data in a field of plots of null data where any patterns seen are due to sampling variability. The Rorschach protocol, from the same paper, shows only null plots to give the reader some intuition for what spurious sampling patterns exists. Savvides et al. (2019) provides a statistical super-test against multiple comparisons driven by probabilistic arguments. There is a CDA quality to these approaches. The sum total of a lot of CDA is not EDA, just as swinging a high-pressure jet around a room is not equivalent to using a misting spray. While EDA and CDA may be along a continuum, we cannot simultaneously perform EDA and CDA as the approaches are, philosophically speaking, perpendicular to one another.

To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations. If we accept this fact, then the only possible source of uncertainty in an uncertainty visualisation system that performs EDA is from the data itself. That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself represents inference that has been done earlier in our analysis. We can see this in the original description of Figure 1 where the uncertainty in all cases, even

measurement error, represents inference that was performed earlier in our analysis.

In the VSUP approach, our ability to arbitrarily decide which values to blend at, or which suppression approach to use, means that the "uncertainty" we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself. This makes the approach antithetical to EDA. For a visualisation to be suitable for EDA, it should always look the same regardless of what hypothesis we plan to draw from it. As much of the transparency in data visualisation comes from this feature in EDA, it is reasonable to set it as a requirement of our visualisations. Ensuring this property means we cannot treat signal and noise as separate variables, but rather as a single integrated unit, the yin and yang of our analysis.

3.4 Uncertainty as a distribution

Rather than trying to summarise our random variables down into a single value or pair of values, why don't we just visualise the whole distribution? This is also a common sentiment in the literature, and we can see the approach in computer science with the HOPs (Hullman et al. 2015), in geoscience with pixel-maps (Lucchesi et al. 2021, Blenkinsop et al. 2000), and in statistics to show multiple forecasts (Hyndman & Athanasopoulos 2021).

3.4.1 Example: visualising samples

Figure 3 shows the six examples introduced in Figure 1, shown as distributions. The distribution us represented by either a sample of outcomes or a quantile dot plot. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. The distribution used for both the slope and intercept is normal with the conventional mean and standard error. Plots S1 and S2 show a pixel map, which is a choropleth map

where each county is coloured by a sample of outcomes from the distribution. Plots G1 and G2 show each univariate distribution as a quantile dot plot. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the univariate distributions are all clearly visible in the low variance case, but disappear in the high variance cases. The upshot is a graphic has achieves signal suppression. Visualising the random variable as a distribution gives additional information, such as the previously hidden bimodality of the univariate distributions in G1 and G2.

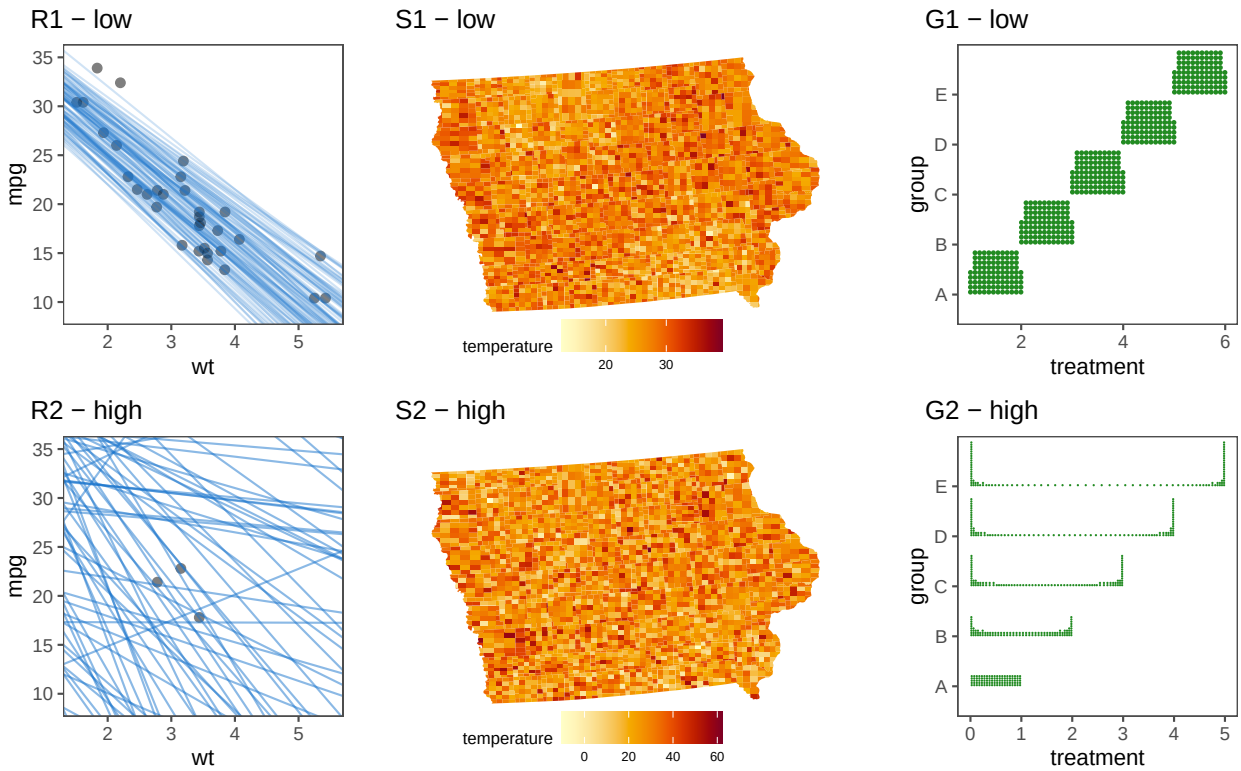

Figure 5: Treating the uncertainty as a distribution, with the same six examples. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. Plots S1 and S2 show a pixel map. Plots G1 and G2 show the groups as quantile dot plots. In each case the signal (regression line, sine wave, increasing trend) has disappeared with high uncertainty.

3.4.2 Quantified versus unquantified uncertainty

By showing our distribution as "data", we are able to read the "uncertainty" plots using the same perceptual tasks that are used to read the "non-uncertainty" plot. This should lead to more effective communication, as people tend to read more complicated visualisations like the bivariate and VSUP plots the same way they read the simple choropleth counterpart (Ndlovu et al. 2023). This approach also does not significantly hinder our ability to extract the individual values mapped by the previous plots, as extracting global statistics from a sample can be done with relative ease (Franconeri 2021). We are able to include more information by offloading more computation to visual processing, but what is the limitation on this approach? Which aspects of uncertainty should be processed by our visual system, and which should be processed by our statistical computation? It is not obvious from the question, but this is actually a question about how much of our uncertainty should be quantified.

Quantified uncertainty usually focuses narrowly on concepts such as probability, confidence intervals, variance, error, or precision (Hullman et al. 2018, Maceachren et al. 2012, Thomson et al. 2005), while unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity (Griethe & Schumann 2006, Wilkinson 2005, Pang et al. 1997, Pham et al. 2009, Boukhelifa et al. 2017). When discussing uncertainty, we typically include these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference. This is often because these unquantified uncertainties violate our assumptions of the uniformity of nature (Otsuka 2023).

Sometimes we are able to visualise these assumption violations directly. For example, we can check for structure in our missing data using the `naniar` package Tierney & Cook (2023)

that allows us to include missing values as a "shadow" alongside our usual visualisations. This approach amounts to just "showing the data", which is a simple but effective option for uncertainty visualisation that is largely overlooked. While this approach is useful for better understanding data, it will not eliminate trends that have become invalid due to structure in missing data or an invalid model. We can only integrate uncertainty as noise when that uncertainty has been *quantified* as an effect on the estimates we are visualising. This is not to say you should do one method over another; visualising both quantified and unquantified uncertainty is necessary for a healthy analysis. Data analysis often works in cycles, where we find assumption violations using EDA, quantify the effect of these violations on our inference, and then visualise the output of that inference using uncertainty visualisation.

4 Evaluating uncertainty visualisations

Unfortunately, the conflicting results in the field are not limited to plot design and extend to the experimental findings as well (MacEachren et al. 2005, Kinkeldey et al. 2014, Hullman 2016). There are as many explanations for the noisy evaluation studies as there are contradictions in the research itself. Kim et al. (2019) believes there is some interference in results from participants' prior beliefs; Hullman (2016) believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or a misunderstanding of statistical concepts (such as confidence intervals). Kinkeldey et al. (2014) suggest the perception of visualisation changes by audience, so we cannot expect the same results between different subpopulations. Brennen & Tuerk (2018) attributes evaluation difficulties to cognitive load from complicated uncertainty visualisations, as well as the participants' prior experience in the topic. While these issues will certainly have some impact on our ability to synthesise, none of them is unique to

uncertainty visualisation. Rather, the issue is likely due to a disconnect between the evaluation methods used and the stated goals of any one experiment.

4.1 Current evaluation methods

4.1.1 Value extraction

Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance (Kinkeldey et al. 2014, Hullman et al. 2019). This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph (Vanderplas et al. 2020). Unfortunately, this approach doesn't work for uncertainty visualisation. The second we ask a specific question about a statistic, that statistic becomes our inferential statistic, even if the plot was not the intent behind the question. By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X} < x)$, we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal suppression. Even if the authors do not realise it themselves, there is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation. Some of the findings are particularly obvious, such as studies that found participants were more accurate when reading a probability expressed as text than when they had to extract it from a graphic (Cheong et al. 2016, Savelli & Joslyn 2013). Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to an area (Ibrekk & Morgan 1987, Gschwandtner et al. 2016), a finding that established the hierarchy of perceptual tasks from more than 40 years ago (see Cleveland & McGill (1984) and replicated by Heer & Bostock (2010)). This extends beyond simple accuracy evaluations, as some authors have found that colour was more

effective than size when searching for maximums and minimums in variances (Sanyal et al. 2009), even though it has been well known that pre-attentive aesthetics, such as colour, are more efficient for search tasks since the 1980s (Vanderplas et al. 2020). By classifying these studies as evaluations of "uncertainty visualisation" while evaluating uncertainty as a signal, we are encouraged to see successful examples of signal suppression as failure. This approach leads authors to advise against particular aesthetic mappings for uncertainty, because they cause participants to have more difficulty extracting values (Blenkinsop et al. 2000). This conclusion is antithetical to the goals of signal suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.

4.1.2 Trust, confidence, and risk aversion

If we cannot directly measure uncertainty for fear that it turns into a signal, we might then assume we can measure the secondary benefits of increased transparency. This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confidence, and risk aversion are all frequently used in uncertainty evaluation studies (Hullman et al. 2019). Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for and against the inclusion of uncertainty.

This is most commonly noticed in the use of trust as a measure, as several authors have commented that measuring trust, and not transparency, can lead to a questionable subtext that argues against transparency (Spiegelhalter 2017, O'Neill 2018). We see this directly play out in the visualisation literature, where surveyed visualisation authors explicitly said they didn't include uncertainty due to the fact that they might decrease trust in their conclusions (Hullman 2020). This sentiment is also true for confidence, as Blenkinsop et al. (2000) commented that visually integrable depictions of uncertainty should be avoided, as

they decrease the viewer's confidence in their extracted data values. Confidence is also used as a proxy for poorly constructed visualisation in standard visualisation experiments, so using confidence as a measure of uncertainty collapses these confounding factors down into a single metric.

Another secondary effect that is similar to trust and confidence is risk aversion. Risk aversion is an economic term used to describe an agent who would choose a random variable with a lower expected payout because it also has a lower variance. Risk aversion's role in the uncertainty visualisation is unclear, as authors will argue uncertainty should elicit more risk aversion in one paper (Hullman et al. 2019), and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next (Wu et al. 2023). Ultimately, these approaches have similar issues to the value extraction studies, except they are slightly more confusing in their goals, leading them to simultaneously argue for and against the inclusion of uncertainty in a visualisation.

4.1.3 Open-ended questions

Some authors try to incorporate uncertainty by asking open-ended questions but compare the results to a ground truth, creating a situation where plots are *evaluated* as a value extraction study without participants being asked the *questions* of a value extraction task. Ibrekk & Morgan (1987) asked participants for the "best estimate" and then compared the result to the distribution's expected value. Hullman et al. (2015) asked participants to compare two normally distributed groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean. Gschwandtner et al. (2016) asked participants the "probability that the interval has already ended at the marked point in time?" and participants replied with the probability that the interval had

already started. The confusion around trying to capture the effects of uncertainty can also extend to the authors of the study itself. We can see an example of this in Padilla et al. (2017), In order to answer the question correctly, the first experiment required participants to assume that an oil rig being"more likely to be hit" by a hurricane would *not* translate to the rig sustaining "more damage". The second experiment required participants to assume the opposite.

4.1.4 Effective methods

This is not to say all evaluation studies fail to properly evaluate uncertainty as noise. There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate (Kale et al. 2018, Correll & Gleicher 2014), which seems to be an effective method. The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* (Wilkinson 2005, Wickham 2010) so we regularly see comparisons between disparate plots that would never be considered substitutes for one another outside the artificial "uncertainty visualisation" framework they are placed within. For example, Kale et al. (2018) compared static bar charts with error bars to a bar chart with animated samples, meaning that any difference in participants' ability to read the plot could be due to the statistic (confidence interval versus sample), the geometry (bars versus intervals), or the use of animation (static versus animated plots). This means that even when evaluations *are* done correctly, there is no generalisable theory we can take from the results.

The point here is not to mock the existing literature for poor academic rigour. If you were to go read these papers yourself, you would find them to (usually) be logically consistent and well-formulated pieces of work. Rather, the point is to illustrate that evaluating

uncertainty as noise is surprisingly difficult. Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

4.2 Implicit Hypothesis Testing

The main problems with current uncertainty visualisation evaluations are that they often require explicit (or convoluted) questions about the variance. Asking direct questions about the statistics or outcomes is not an explicit requirement of visualisation evaluations. In their review of testing statistical graphics, Vanderplas et al. (2020) drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot. The lineup protocol is the most salient example of the implicit approach. Lineups are a confirmatory visualisation tool where participants are shown a set of $M - 1$ plots, and asked to identify the plot that is the "most different", leaving participants to decide what "most different" means to them, even if it is not what the authors intended (VanderPlas & Hofmann n.d.). The implicit testing does not limit the versatility of the approach, with the lineup being used to evaluate the effectiveness of different types of plots (Hofmann et al. 2012), colour palettes (Reda & Szafir 2021), and design decisions (VanderPlas & Hofmann n.d.).

Lineup protocols are not only useful for their implicit testing, but they also have parallels to hypothesis testing that can be leveraged for uncertainty visualisation. The concept of signal suppression is, at its core, an assertion of statistical validity, which means the visibility of signals should be directly proportional to p-values or some equivalent measure. This comparison is not new in uncertainty visualisation, where parallels have been drawn to

frequentist statistics by Correll & Gleicher (2014), who compared results to Cohen's D, and to Bayesian statistics by Kim et al. (2019), who evaluated plots based on their impact on users' prior beliefs. The comparison to hypothesis testing is far more natural for the lineup protocol, with Majumder et al. (2013) explicitly defining a visual test statistic, while Li et al. (2024) compared lineups to standard statistical tests using power curves. The parallels to lineups and uncertainty visualisation are numerous, with the relationship even being identified by Hullman et al. (2015) in their development of hypothetical outcome plots.

The lineup protocol and uncertainty visualisation share a lot in common, with the motivation behind lineups also rooted in checking if perceived patterns are real or merely the result of chance (Buja et al. 2009, Wickham et al. 2010). As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the lineup protocol directly to evaluate uncertainty visualisation, but a new evaluation methodology should try to learn from the success of the lineup approach. Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance would solve many of the issues with the current evaluation approaches.

4.3 Heuristics research

There is one element of uncertainty visualisation research that we have not touched on yet, but it is incredibly consistent, and that is the research around visual heuristics. The heuristics research, unlike the plot evaluation work, is incredibly consistent, likely because heuristics are indifferent to the goal of our analysis and uncertainty's role as signal or noise. Several heuristics are of particular importance for uncertainty visualisation. We will give a brief overview of them here, and for a more in-depth analysis, see Padilla, Kay & Hullman (2022).

The use of confidence intervals or boundaries has a few known issues in their interpretation. The sine illusion can cause the confidence interval of a smoothed sine curve to seem wider at the peaks than the troughs, causing us to underestimate uncertainty associated with changing values (Vanderplas & Hofmann 2015). Similarly, researchers frequently find that visual boundaries such as the cone of uncertainty in hurricane forecasts (Padilla, Kay & Hullman 2022), or confidence bands or ellipses (VanderPlas & Hofmann n.d.) create the impression of a meaningful bound, separating points outside the bounds from those within it, which can create problems if said bound is not inherently meaningful.

While many researchers tend to prefer frequency framing, they are not without their issues. Points that were on an outcome of an ensemble display were perceived as more likely than points not on an outcome, even when the point that was not on an outcome was closer to the centre of the distribution (and therefore more likely) (Padilla et al. 2017). This is considered to be a variation of the within-bar bias, where outcomes within the bar on a bar chart as perceived to be more likely than those outside it (Newman & Scholl 2012).

We should also try to avoid visualising central estimates if possible, as several studies have found that viewers ignore uncertainty information and compare the difference between the two means if they have the option (Padilla, Kay & Hullman 2022).

These heuristics have the potential to create noise in signal suppression evaluations, and should be kept in mind when designing uncertainty visualisations as well as evaluation experiments.

5 Conclusions and Future Work

This paper has summarised the literature and provides suggestions for structural framework around uncertainty visualisation. Particularly, we propose that the purpose of an uncertainty visualisation be signal-suppression, to dampen weak signals and amplify strong signals. It has highlighted several gaps in the existing literature.

Experimental practices on uncertainty visualisation need standards. Some existing evaluation experiments treat uncertainty as a signal, while others treat uncertainty as noise. As the literature currently exists, it is difficult to combine results from papers to get a meaningful sense of how uncertainty information is understood by a viewer. Researchers need to ensure that when they identify the motivation behind their visualisation technique, their evaluation methods align with the stated goals of the paper.

Experimental methods that evaluate uncertainty as noise need to be developed. Research into separability and integrability of signal and noise is of particular interest to uncertainty visualisation, as it allows assessment of the interference between the two. When designing experiments, authors often choose aesthetics that are visually distinguishable; uncertainty visualisation authors should consider doing the opposite.

Uncertainty needs to be formalised within the grammar of graphics. Some of this formalisation was done by Kay (2023), but it focuses only on the visualisation of univariate distributions. Giving authors the ability to describe uncertainty visualisations in terms of statistics, geometries and aesthetics will support evaluation experiments that can build towards a cohesive theory of visualising uncertainty.

Software that allows users to easily perform signal suppression is needed. Existing uncertainty visualisation methods view a distribution as its own object, and there are no software options

treating "an uncertainty visualisation as a function of an existing visualisation" philosophy.

Signal suppression is an undeveloped area of visualisation research, and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation.

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